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Final Project Review

DISEASE ANALYSIS USING HOSPITAL DATA

BATCH NUMBER: 15

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Problem Formulation

- In today's healthcare systems, large volumes of hospital data such as patient records, disease details, age, gender, and medical information are generated daily. Effectively analyzing this data to identify disease patterns and trends remains a significant challenge.
- Existing healthcare analysis systems often rely on manual methods or basic statistical techniques, which may fail to:
- Accurately identify frequently occurring diseases
- Analyze disease distribution across different age groups and genders
- Handle large-scale and diverse hospital datasets
- Provide meaningful insights for better healthcare decision-making
- Maintain accuracy and consistency in analysis results

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Problem Definition

- Traditional methods of analyzing hospital data rely on manual analysis and basic statistical techniques, which are not efficient for handling large datasets.
- These existing methods have several limitations:
- Unable to efficiently handle large-scale hospital datasets
- Fail to identify frequently occurring diseases accurately
- Lack the ability to analyze disease patterns based on age and gender
- Do not provide real-time and meaningful insights for decision-making
- Results are often less reliable and time-consuming

Literature Review

Paper Title	Journal Details & Authors	Methodology	Limitations
Predicting 30-Day Hospital Readmission for Heart Failure Patients	Journal of Biomedical Informatics, 2023 – Miller et al.	Utilizes ensemble learning techniques, specifically **XGBoost and LightGBM**, on electronic health record (EHR) data. Performs extensive feature engineering on patient history and physiological data.	Accuracy is affected by variations in data recording practices across different departments. Model lacks interoperability without additional explainability tools.
Operational Efficiency and Resource Allocation via Real-time ICU Bed Demand Forecasting	Health Care Management Science, 2024 – Zhao et al.	Implements **LSTM (Long Short-Term Memory)** recurrent neural networks on historical admission and patient flow data to forecast real-time bed occupancy.	Highly sensitive to data quality and sudden outliers (e.g., localized outbreaks). Model requires continuous training on new data streams.
Uncovering Patient Phenotypes for Personalized Treatment Plans	Data Mining and Knowledge Discovery, 2022 – Rossi et al.	Applies unsupervised machine learning, including **K-Means Clustering and PCA**, to group diverse hospital patients based on complex multi-morbidity patterns.	Clustering results can be difficult for clinicians to interpret and integrate into standard care pathways. Scalability on extremely large, high-dimensional datasets.
Federated Learning for Privacy-Preserving Mining of Distributed Hospital Data	IEEE Transactions on Cloud Computing, 2025 – Ibrahim et al.	Proposes a decentralized framework using **Federated Learning** to train machine learning models across multiple hospitals without raw data sharing.	Significant communication overhead between nodes. Vulnerable to potential model poisoning attacks if participant security is compromised.

Research Gap:

Existing systems for healthcare data analysis mainly rely on traditional statistical methods or use individual data mining techniques without proper integration. Many approaches focus only on limited patient attributes and do not efficiently handle large, real-world hospital datasets. Additionally, data preprocessing and feature selection are often not properly implemented, which affects the accuracy and reliability of the analysis.

Therefore, there is a need for an integrated data mining system that combines data preprocessing, feature selection, and analytical techniques to provide accurate, scalable, and reliable insights into disease patterns for effective healthcare decision-making.

Objectives:

- To develop a data mining-based system for analyzing hospital data and identifying disease patterns
- To perform effective data preprocessing and cleaning to handle missing values, noise, and inconsistencies in the dataset
- To analyze patient information such as age, gender, and disease type to identify frequently occurring diseases
- To apply data mining techniques such as classification (Decision Tree, Naive Bayes) and clustering (K-Means) for disease analysis
- To visualize disease distribution and patterns using charts and graphs
- To improve the accuracy and reliability of analysis through proper data processing and evaluation methods

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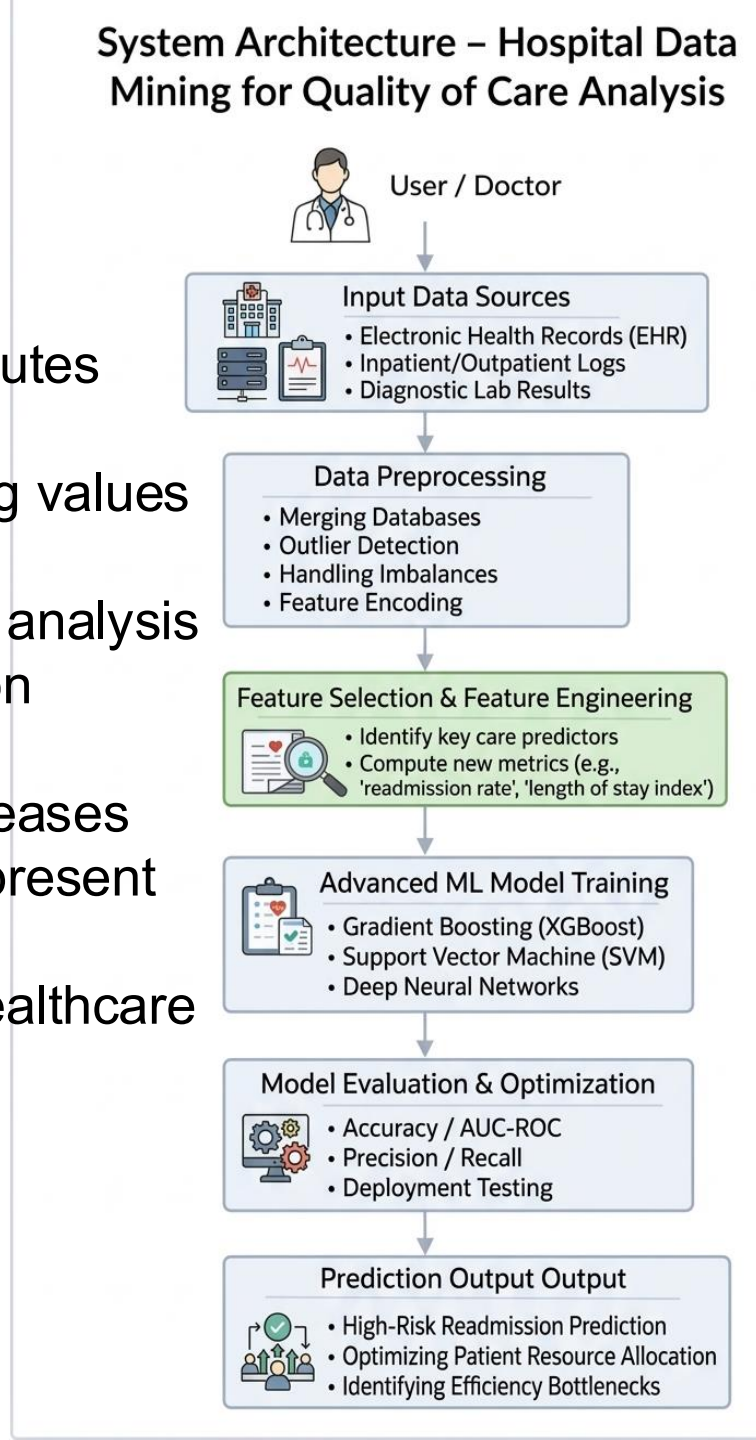
Challenges and Risks:

- Handling large and complex hospital datasets with incomplete or inconsistent records
- Managing missing values and noisy data in patient information
- Identifying important attributes such as age, gender, and disease type for accurate analysis
- Ensuring proper data preprocessing to avoid incorrect or misleading results
- Maintaining accuracy and consistency in disease pattern analysis
- Ensuring data privacy and security of patient records
- Handling real-world data variations across different patient groups

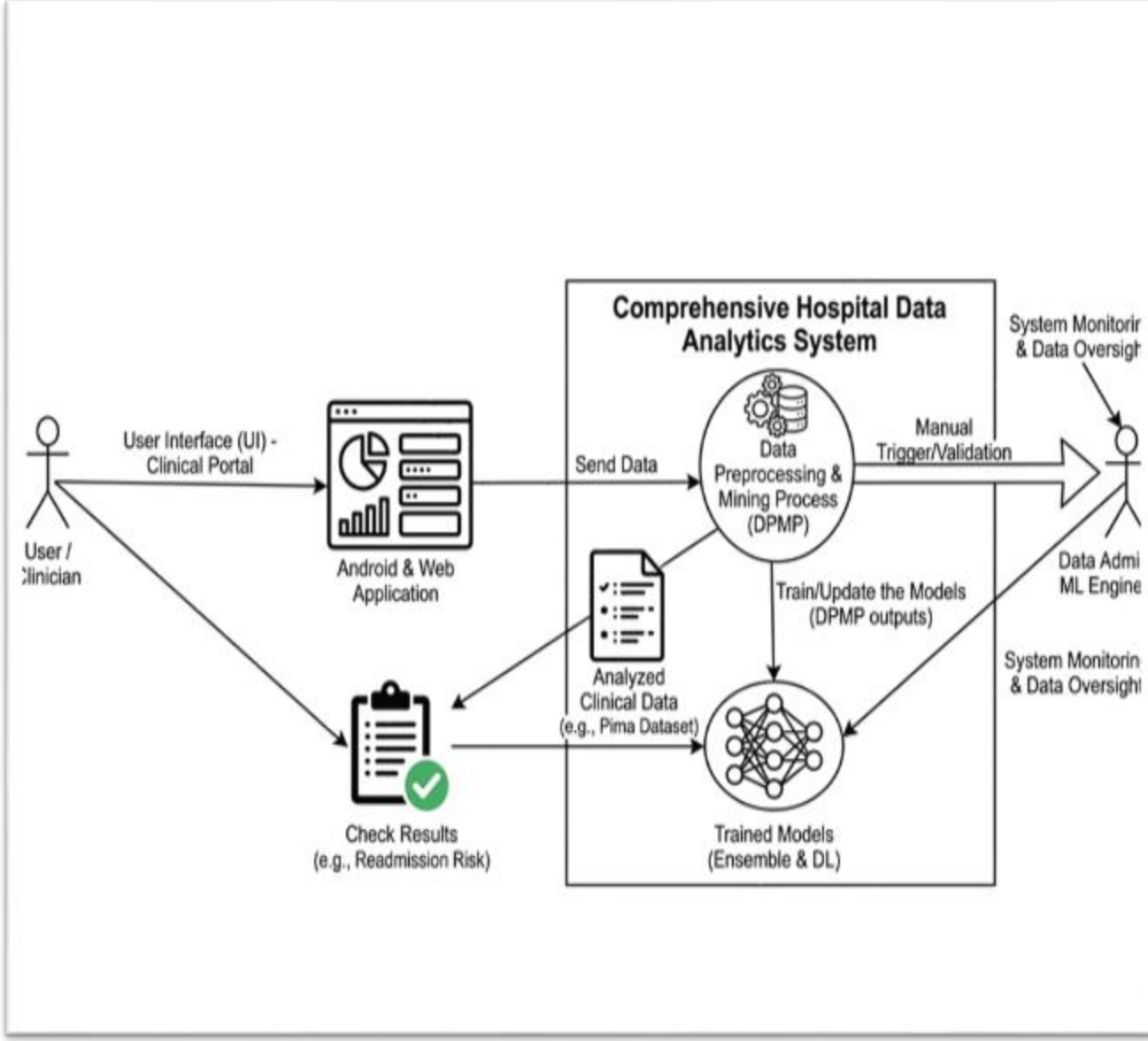
Methodology

- Hospital patient data is collected from a dataset (CSV file) containing attributes such as Patient ID, Age, Gender, and Disease
- Data preprocessing is performed, including data cleaning, handling missing values removal of duplicates, and standardization of data
- Important features such as age, gender, and disease type are selected for analysis
- The dataset is analyzed using data mining techniques such as classification (Decision Tree, Naive Bayes) and clustering (K-Means)
- Disease frequency analysis is performed to identify the most common diseases
- Visualization techniques such as bar charts and pie charts are used to represent disease distribution and patterns
- The final results are analyzed to understand disease trends and support healthcare decision-making



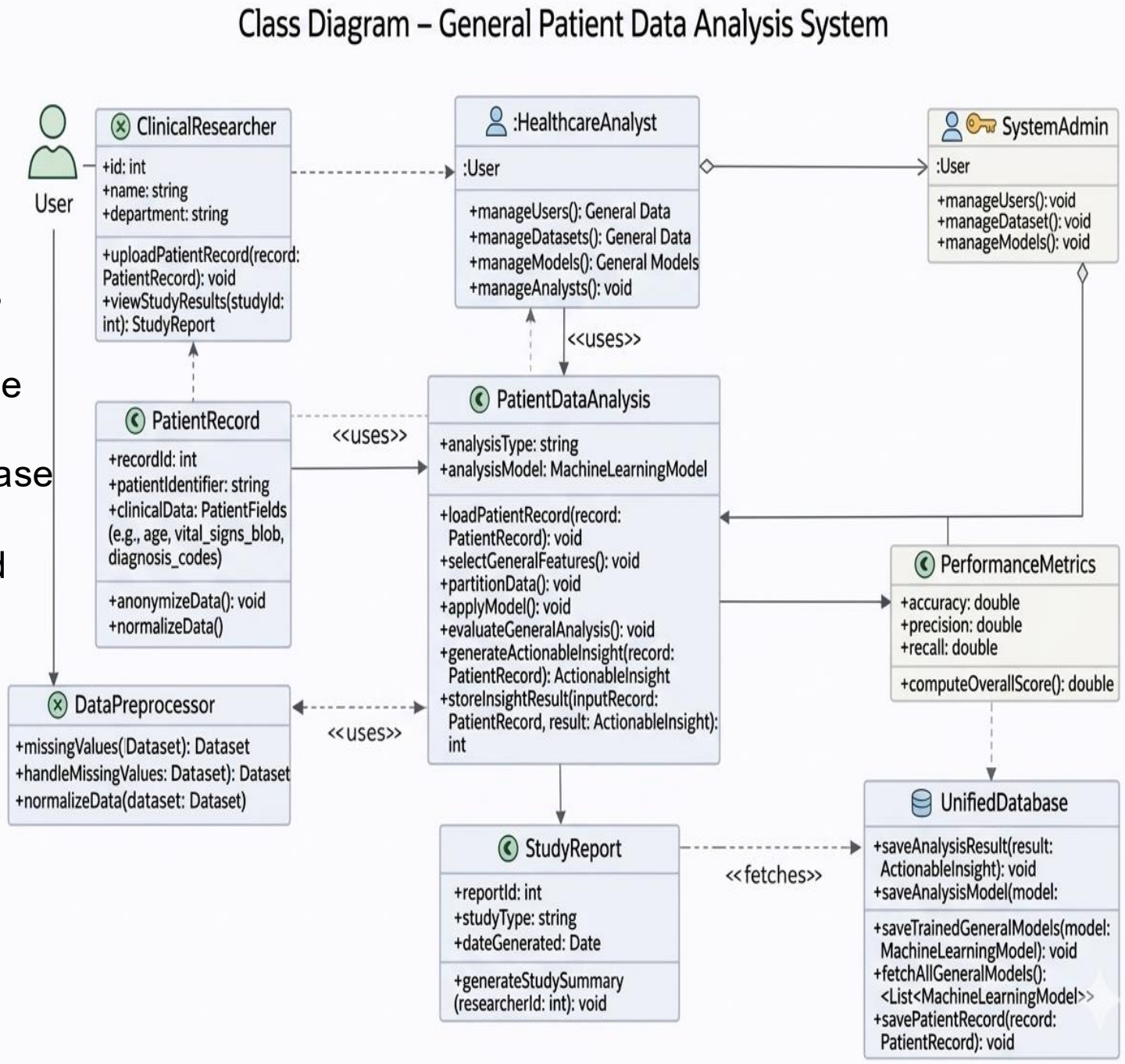
Use Case Diagram

- User uploads hospital dataset and views analysis results
- System performs data preprocessing and disease analysis
- Data mining techniques are applied to identify disease patterns
- Results are displayed and can be downloaded or exported



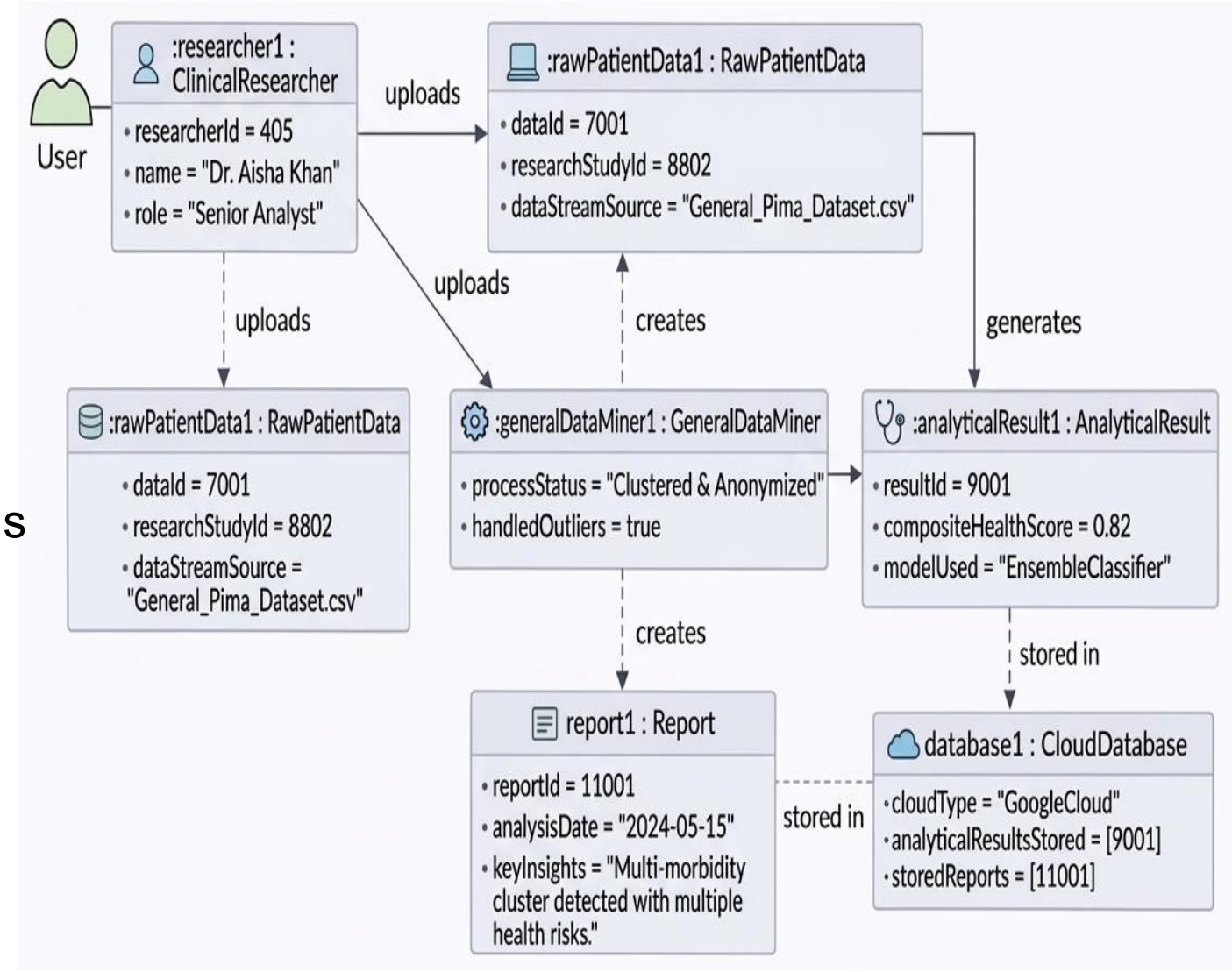
Class Diagram

- The user (hospital staff) uploads patient data and requests disease analysis; the system processes the data and displays results
- The system manages data preprocessing, disease analysis, and storage of results
- Data mining techniques are used to analyze disease patterns and generate insights
- The interface/dashboard allows users to view and export analysis reports



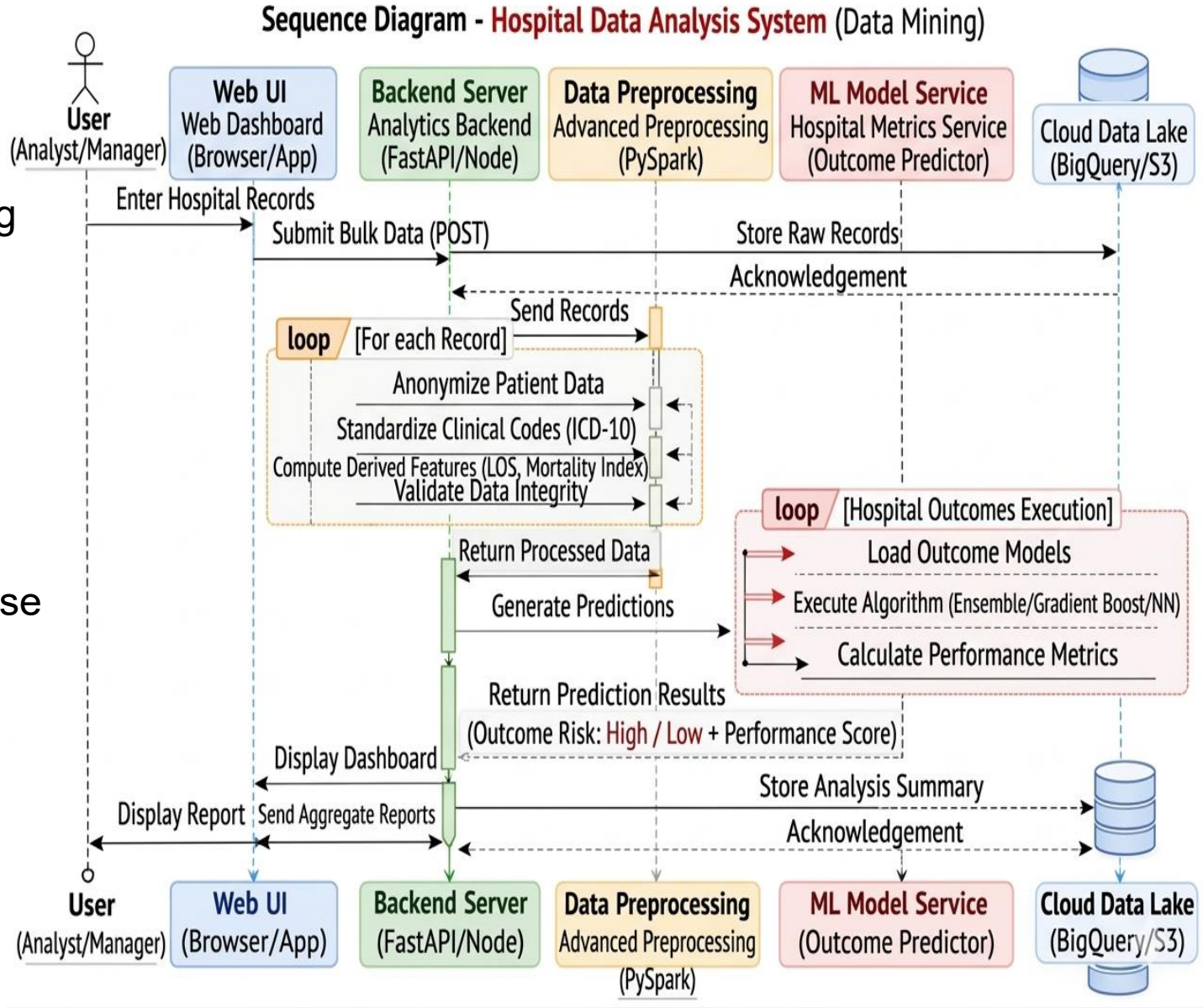
Object Diagram

- Shows objects like user, dataset, and analysis results during execution
- Represents how patient data is stored and processed
- Displays relationships between dataset, analysis results, and reports



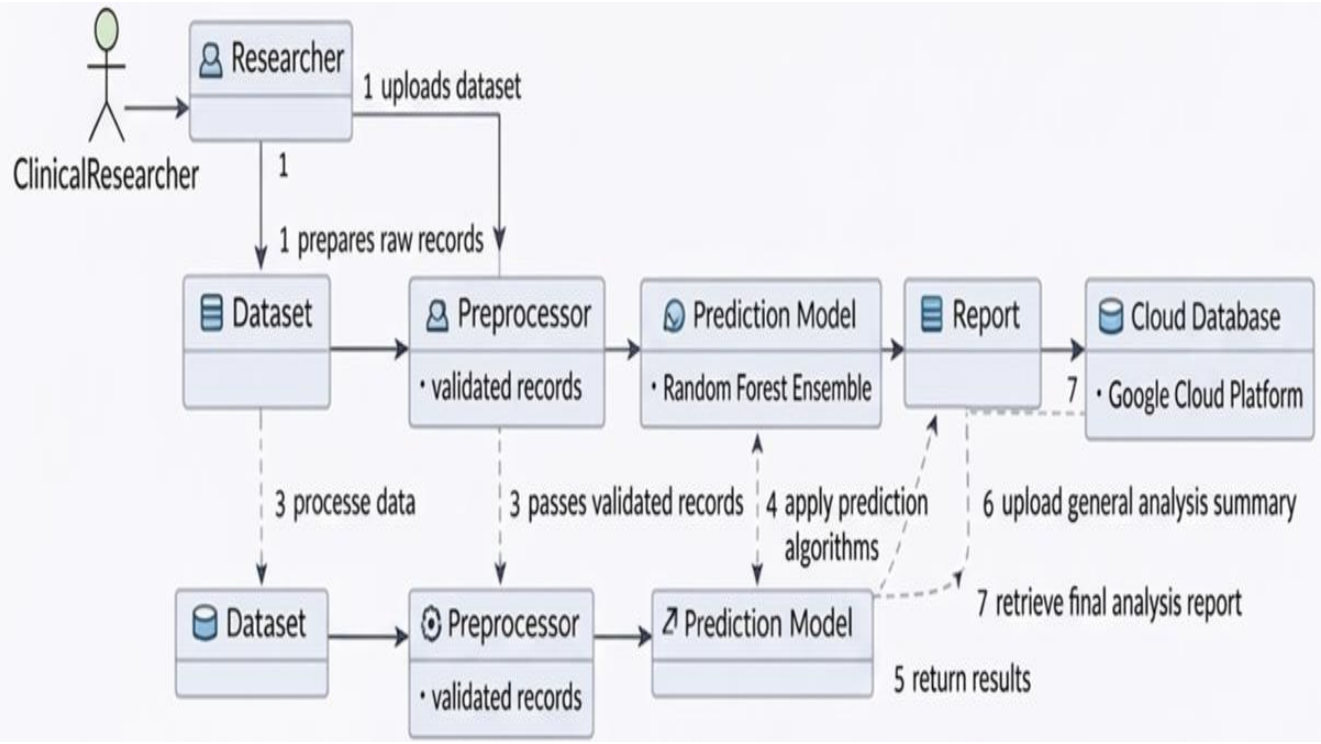
Sequence Diagram

- Step-by-step flow of disease analysis using hospital data
- User uploads patient dataset → data preprocessing → disease analysis using data mining techniques → results generation → storage
- Analysis results are displayed on the dashboard
- Ends with report generation and summary of disease distribution and insights



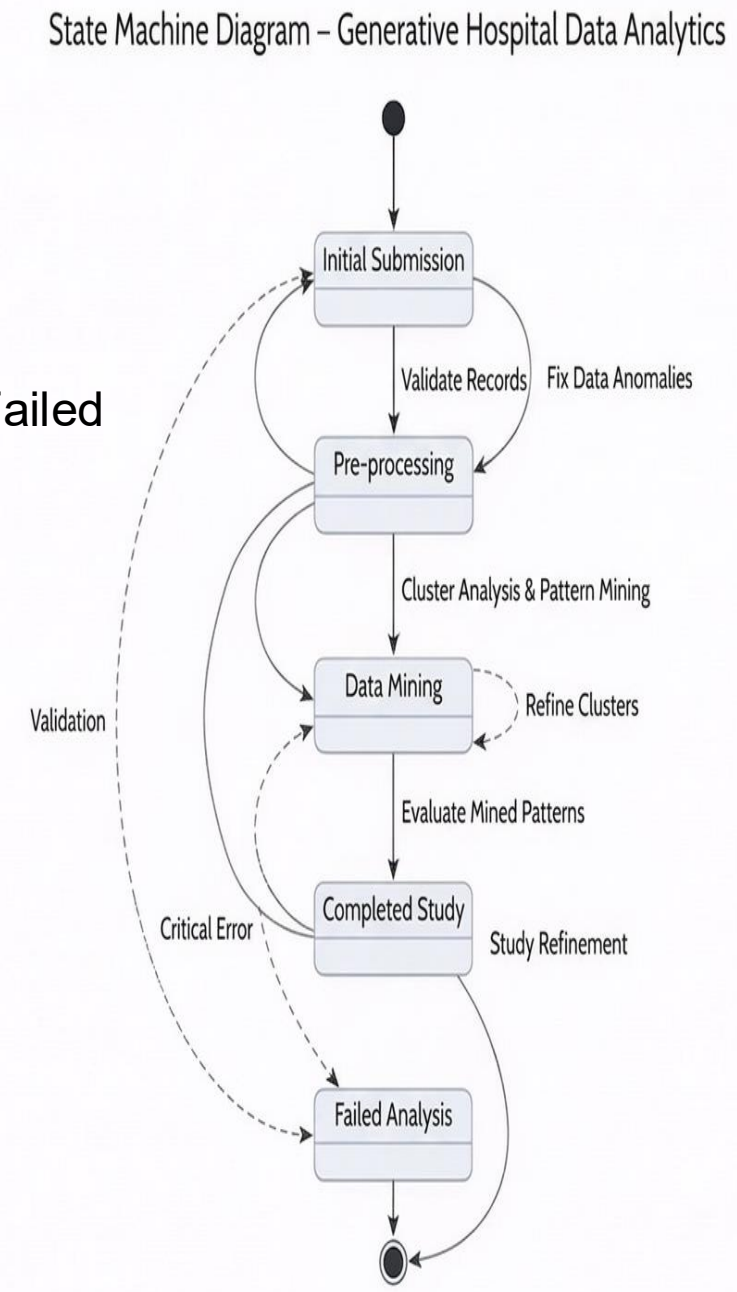
Collaboration Diagram

- Shows communication flow between system components
- Represents steps like dataset upload → preprocessing → analysis → result generation
- Displays how results are sent to the user for viewing and download



State Machine Diagram

- ➢Data Uploaded → Preprocessing → Disease Analysis → Result Generated / Failed
- ➢The process repeats for each patient record or dataset entry
- ➢Ends when analysis results are successfully generated or if an error occurs



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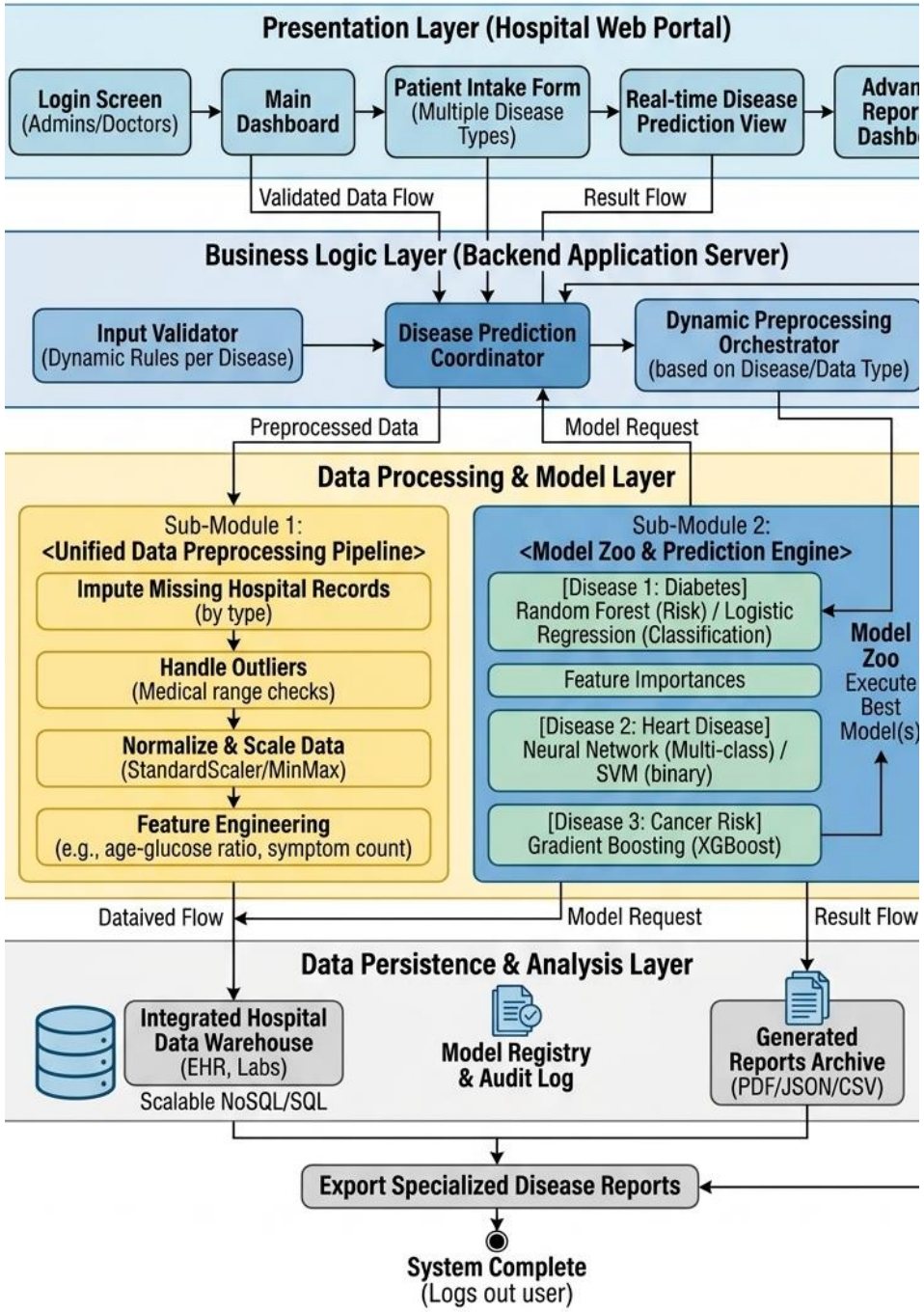
Activity Diagram

- ➢Represents the end-to-end workflow of the disease analysis process.
- ➢Hospital data entry → data preprocessing → feature analysis → application of data mining techniques
- ➢Analysis results are generated → stored → dashboard/report updated
- ➢Final output includes disease patterns and analysis summary
- ➢Allows exporting analysis reports for further use



Deployment Diagram

- Shows how the system operates across different components and environments.
- Client interface (User/Hospital Staff) sends hospital dataset to the backend system
- Backend processes data using data mining techniques for disease analysis
- Server performs preprocessing, analysis, and clustering operations
- Database or cloud storage manages patient records and analysis results

Technology Requirements

Software Requirements

- **Programming Language:** Python 3.x
  Used for data preprocessing, analysis, and implementation of data mining techniques
- **Data Mining / Machine Learning:**
  Used to analyze hospital data and identify disease patterns using techniques like classification and clusterin
- **Tools & Frameworks:** WEKA / Jupyter Notebook
  Used for applying data mining algorithms and performing analysis
- **Database:** MySQL / PostgreSQL
  Used to store hospital datasets, processed data, and analysis results
- **Frontend (Optional):** HTML, CSS, JavaScript
  Used to display input data and visualization results (charts/reports)

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Hardware Requirements

- **Processing Unit:** Minimum Intel Core i5 (8th Gen) / AMD Ryzen 5
  Required for data processing and model training
- **Memory (RAM):** Minimum 8 GB
  Supports handling datasets and ML models
- **Storage:** Minimum 256 GB SSD
  Stores datasets, trained models, and prediction records
- **Input Devices:** Keyboard, Mouse, and Internet Connection
  Required for data entry and web access

Dataset Requirements:

HospitalDataset(CSVFile):

➢Includes attributes such as Patient ID, Patient Name, Age, Gender, and Disease

ExtendedMedicalDataset(Optional):

➢May include additional attributes such as Blood Pressure, Sugar Level, Cholesterol, and Admission Type for deeper analysis

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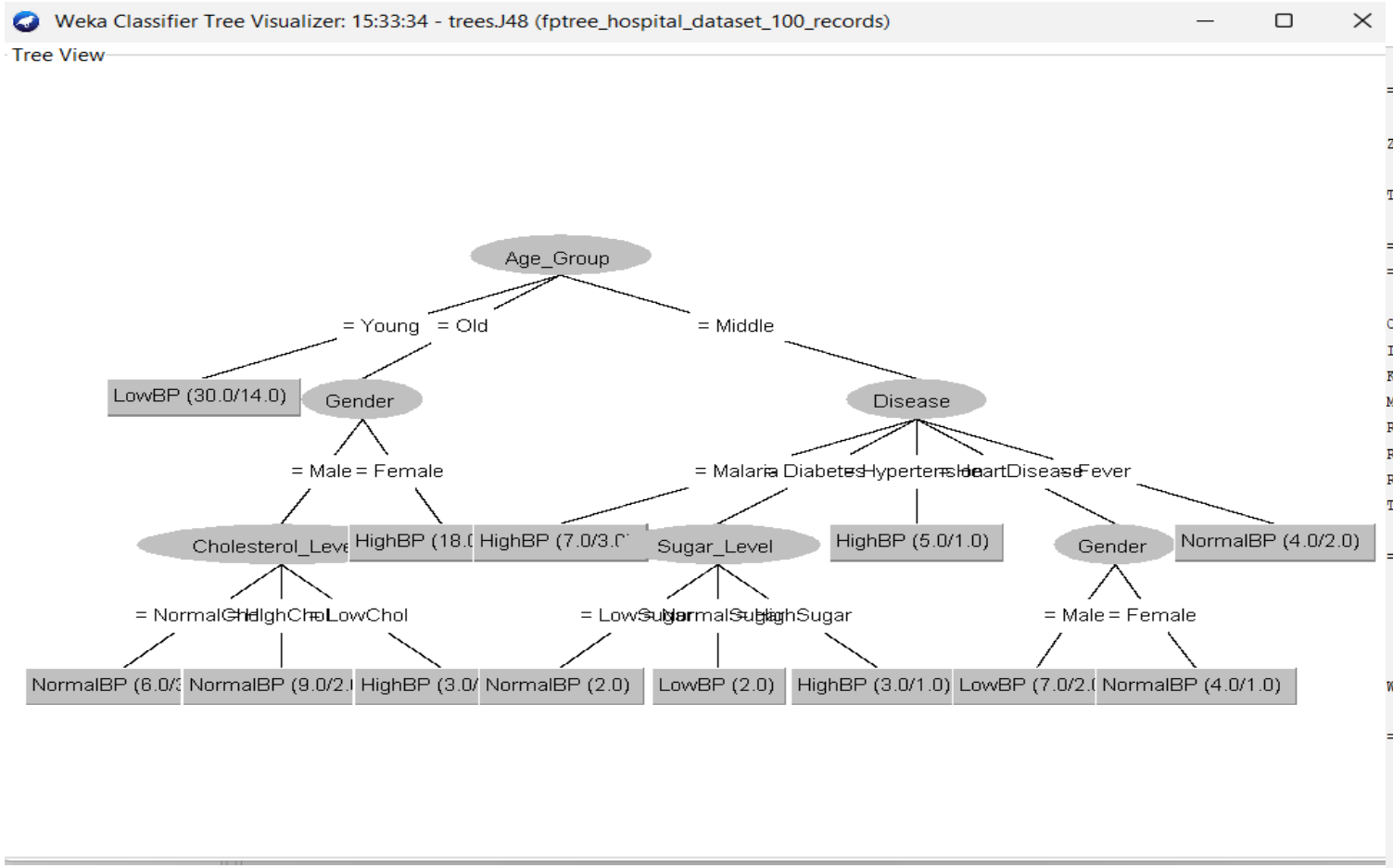
Final Result




```
=== Classifier model (full training set) ===

ZeroR predicts class value: Regular

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances          68               68      %
Incorrectly Classified Instances        32               32      %
Kappa statistic                          0
Mean absolute error                      0.437
Root mean squared error                  0.4669
Relative absolute error                100      %
Root relative squared error            100      %
Total Number of Instances              100

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.000    0.000    ?          0.000    ?          ?        0.463     0.305     Emergenc
                 1.000    1.000    0.680      1.000    0.810      ?        0.463     0.666     Regular
Weighted Avg.    0.680    0.680    ?          0.680    ?          ?        0.463     0.550

=== Confusion Matrix ===

  a  b   <-- classified as
  0 32 |  a = Emergency
  0 68 |  b = Regular
```

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Conclusion

- The system successfully analyzes hospital data using data mining techniques to identify frequently occurring diseases and their patterns. Effective data preprocessing and analytical methods improve the accuracy and reliability of the results, making the system useful for healthcare analysis and decision-making.

- **Future Enhancements**
- Integration with real-time hospital systems for continuous data analysis
- Development of a web or mobile application for easy access and monitoring
- Implementation of advanced data mining techniques for improved analysis
- Addition of predictive features to forecast disease trends

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- **References**

- [1] A. Kumar and R. Sharma, “Data Mining Techniques for Disease Analysis Using Healthcare Data,” *IEEE Access*, vol. 9, pp. 11234–11245, 2021.
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- [5] F. Chollet, *Deep Learning with Python*. Manning Publications, 2018.

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